

D07. SQL Operations Pack

Overview

This pack documents the SQL operations used to query, validate, and safely demonstrate a small data modification on the FlowCore Oracle dataset.

The script is intentionally simple and readable. It is designed to run after the schema and sample data files

have already loaded the baseline inbound warehouse scenario.

Scenario Baseline

The SQL operations use the current FlowCore inbound scenario:

- **one inbound receipt: RCV-1001**
- **one dock: DOCK-01**
- **three stored pallets: PLT-1001, PLT-1002, PLT-1003**
- **five warehouse locations: A01, A02, A03, B01, C01**
- **nine pallet lifecycle events**
- **final pallet flow: RECEIVED -> ASSIGNED -> STORED**

Files Used

- **project/sql/MySchema.sql**
- **project/sql/Sample_Data.sql**
- **project/sql/SQL_Operations.sql**

SQL Coverage

The script demonstrates:

- **basic retrieval from project tables**
- **filtering with WHERE**
- **sorting with ORDER BY**
- **joins between related tables**
- **aggregation with COUNT and SUM**
- **validation queries**
- **a temporary UPDATE followed by ROLLBACK**

Retrieval Queries

The first queries confirm that the baseline data exists.

Expected output:

- **RECEIPT** returns one row: RCV-1001, DOCK-01, COMPLETE.
- **PALLET** returns three rows: PLT-1001, PLT-1002, and PLT-1003.

Filtering And Sorting

The filtering section shows how the dataset can be narrowed down.

Expected output:

- The **CURRENT_STATUS = 'STORED'** query returns all three pallets.
- The stored pallet result is ordered by quantity descending: PLT-1003 with 15, PLT-1001 with 12, and
- PLT-1002 with 8.
- The active location query returns four active locations: A01, A02, A03, and B01.

Join Queries

The join section proves that the schema relationships are working.

Expected output:

- The pallet-to-receipt join returns three rows, one for each pallet under receipt RCV-1001.
- The pallet-to-location join returns three rows and confirms the final location of each pallet.
- The status-event-to-pallet join returns nine rows and confirms that every event belongs to a real pallet.

Aggregation Queries

The aggregation section summarizes the operational data.

Expected output:

- **TOTAL_PALLETS = 3.**
- Receipt RCV-1001 has **PALLET_COUNT = 3** and **TOTAL_QUANTITY = 35.**
- Location groups show one pallet each in A01, A02, and A03.
- Status event totals are 3 RECEIVED, 3 ASSIGNED, and 3 STORED.
- The **HAVING** query returns only locations that currently have at least one pallet.

Validation Queries

The validation section is expected to return no problem rows.

Expected output:

- **Stored pallets without a location: no rows selected.**
- **Locations over capacity: no rows selected.**
- **Status events with a missing pallet reference: no rows selected.**
- **This is the correct result. In this project, a validation query returning no rows means that the specific data-**

- **quality problem was not found.**

Temporary Update And Rollback

The final section demonstrates a safe data modification.

Expected output:

- **Before the update, location B01 has priority 1.**
- **The UPDATE statement changes B01 priority from 1 to 2.**
- **Oracle reports 1 row updated.**
- **The next SELECT shows B01 priority as 2.**
- **ROLLBACK cancels the temporary change.**
- **Oracle reports Rollback complete.**
- **The final SELECT shows B01 priority restored to 1.**
- **This proves that the script can demonstrate an update without permanently changing the baseline dataset.**

What The Script Returns

The script returns both data result sets and diagnostic outputs.

The expected outputs are:

- **Receipt check: 1 receipt row.**
- **Pallet check: 3 pallet rows.**
- **Stored pallet filter: 3 rows.**
- **Active location filter: 4 rows.**
- **Pallet-to-receipt join: 3 rows.**
- **Pallet-to-location join: 3 rows.**
- **Status-event join: 9 rows.**
- **Total pallet count: 3.**
- **Receipt quantity total: 35 units.**
- **Event counts by status: 3 rows, one for each lifecycle status.**
- **Validation queries: no rows selected.**
- **Temporary update: 1 row updated.**
- **Rollback: rollback complete.**
- **Final priority check: B01 returns to priority 1.**

These outputs are correct for the current FlowCore baseline because the project intentionally contains three pallets, nine lifecycle events, and no validation violations.

Execution Order

Run the files in this order:

1. **project/sql/MySchema.sql**
2. **project/sql/Sample_Data.sql**
3. **project/sql/SQL_Operations.sql**

The files should be run as full scripts, not as isolated selected lines.